

Empirical Research in Finance

Combined Course Submission · Group A

Humboldt-Universität zu Berlin
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PART I · GROUP SUBMISSION · EMPIRICAL ASSIGNMENT

Firm Valuation and Future Stock Returns

Evidence on the Value Effect and the Brexit Shock
among German Listed Firms

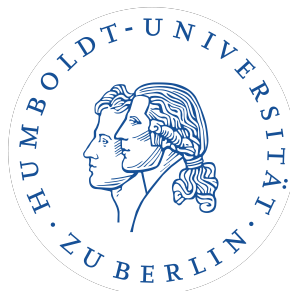
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Abstract

We examine valuation and one-year-ahead excess stock returns in a simulated panel of German listed firms. Higher book-to-market ratios are associated with higher returns, consistent with the Fama–French value pattern, and this relationship is stronger among small firms. It remains positive after including year and industry controls and clustering standard errors by firm. In the accompanying Brexit event study, UK sales exposure identifies substantially more negative abnormal returns, whereas UK listing status does not distinguish the groups’ reactions.

Data

The main analysis uses the simulated annual panel of German listed firms supplied through Moodle for this assignment. The file `value_effect_panel.dta` contains 3,060 firm-year observations for 255 firms from 2010 to 2021, linking characteristics in year t to excess returns in $t + 1$. This panel allows us to examine the value effect across firms and over time. The accompanying `cars_summary.dta` is used later for the Brexit event study and provides group-level abnormal returns and standard errors.

We clean the main panel by excluding financial firms (two-digit SIC 60–69), whose balance sheets and leverage differ from those of non-financial firms. We also require complete identifiers and regression inputs, meaning that the firm and year are known and none of the values needed to estimate the models is missing. The final sample contains 2,664 observations for 230 firms. It is an unbalanced panel because some firms do not have an observation in every year. Table 1 documents the filters.

Table 1. Sample selection

Sample restriction	Removed	Remaining
Original panel	–	3,060
Exclude financial firms (SIC 60–69)	300	2,760
Positive share price	10	2,750
Complete identifiers and regression inputs	86	2,664

Notes. Filters are applied sequentially. Each exclusion is counted once. Positive assets and shares ensure valid ratios. Neither check removes observations. Positive share prices also ensure valid log size. Complete inputs exclude missing values needed for identification or regression. Winsorisation removes no observations.

To limit the influence of extreme values, we winsorise excess returns, book-to-market, leverage, profitability and dividend yield at 1% in each tail across the final sample. This replaces the most extreme values with the respective boundary values without removing observations. Market capitalisation retains its original values to preserve the plausible size distribution. Its natural logarithm, our size measure, already compresses large differences and receives no further tail adjustment. Table 2 summarises the adjusted final sample used throughout the regressions.

Table 2. Final-sample descriptive statistics

Variable	Mean	SD	Min	P10	Median	P90	Max
Excess return	15.683	25.410	-44.288	-17.484	15.523	49.208	74.637
Book-to-market	1.409	0.704	0.100	0.452	1.400	2.357	2.898
Leverage	27.958	16.489	0.000	7.205	26.360	51.382	67.186
Profitability	7.549	7.540	-10.608	-2.217	7.523	17.533	25.816
Dividend yield	1.806	1.592	0.000	0.000	1.679	4.076	5.520
Size	3.981	1.839	-0.899	1.817	3.805	6.426	10.106
Market cap.	324.475	1,181.931	0.407	6.156	44.917	617.959	24,491.312

Notes. The sample contains 2,664 observations for 230 firms. Returns at $t + 1$, leverage, profitability and dividend yield are in percent. Book-to-market is a ratio. Market capitalisation is in EUR million. Size is its natural logarithm. The first five variables are winsorised at 1% in each tail. SD is the standard deviation. P10 and P90 are percentiles.

Annual excess returns average 15.683% and range from -44.288% to 74.637%, a wide but economically possible range for individual stocks. Book-to-market ranges from 0.100 to 2.898, covering valuations above and below book equity. Leverage of 0.000–67.186%, profitability of -10.608% to 25.816%, and dividend yields of 0.000–5.520% are plausible ranges that include loss-making and non-dividend-paying firms.

Market capitalisation ranges from EUR 0.407 million to EUR 24.491 billion. Its mean of EUR 324.475 million exceeds the EUR 44.917 million median, showing that a few very large firms pull the average well above the typical firm. This substantial right skew motivates the size controls used below. Log size ranges from -0.899 to 10.106. Negative log values simply indicate market capitalisation below EUR 1 million.

Methodology and Results

We begin with the baseline model in task (b), using one-year-ahead excess return as the dependent variable and book-to-market as the sole explanatory variable. Pooled OLS combines all firm-year observations in one regression. Every specification uses the same 2,664 observations, so changes in the results reflect changes in the model rather than the sample. The baseline book-to-market coefficient is positive and significantly different from zero (0.088, $t = 12.920$, $p < 0.001$). A one-unit higher ratio is associated with an 8.762-percentage-point higher excess return in the following year.

To assess whether this relationship also reflects other firm characteristics, model (c) adds size, leverage, profitability and dividend yield. For firm i in year t , the model is

$$r_{i,t+1}^e = \alpha + \beta_B \text{B2M}_{it} + \beta_S \text{Size}_{it} + \beta_L \text{Lev}_{it} + \beta_P \text{Prof}_{it} + \beta_D \text{DY}_{it} + \varepsilon_{it}. \quad (1)$$

Here, r^e is excess return, B2M is book-to-market, and *Size*, *Lev*, *Prof* and *DY* denote size, leverage, profitability and dividend yield. The intercept is α and ε is the error term. Returns and financial ratios enter in decimal form. The book-to-market coefficient falls to 0.059 but remains significant ($p < 0.001$), suggesting that the baseline relationship partly overlaps with the added characteristics. Size is negatively associated with returns ($p < 0.001$), while profitability is positively associated ($p = 0.036$). Leverage and dividend yield are not individually significant at 5% ($p = 0.532$ and $p = 0.115$). We retain them to account for financing and payout differences when estimating the other coefficients. A non-significant individual test does not establish that a variable is irrelevant as a control. Table 3 consolidates all regression results.

Table 3. One-year-ahead excess returns

	(b)	(c)	(d)/(e)	(f)	(g)
Book-to-market	0.088*** (0.007)	0.059*** (0.008)	0.043*** (0.010)	0.044*** (0.008)	0.044*** (0.010)
Size		-0.023*** (0.003)			
Small firm			0.033 (0.022)	0.034* (0.019)	0.034* (0.020)
Book-to-market $\times$ Small			0.046*** (0.014)	0.045*** (0.012)	0.045*** (0.013)
Leverage		-0.018 (0.029)	-0.025 (0.029)	-0.028 (0.025)	-0.028 (0.024)
Profitability		0.149** (0.071)	0.161** (0.071)	0.131** (0.062)	0.131** (0.056)
Dividend yield		0.529 (0.336)	0.330 (0.336)	0.500* (0.288)	0.500* (0.290)
Constant	0.033*** (0.011)	0.150*** (0.022)	0.031* (0.017)	-0.139*** (0.024)	-0.139*** (0.025)
Observations	2,664	2,664	2,664	2,664	2,664
R^2	0.059	0.085	0.099	0.349	0.349
Adjusted R^2	0.059	0.083	0.097	0.343	0.343
Year dummies	No	No	No	Yes	Yes
Industry dummies	No	No	No	Yes	Yes
Standard errors	OLS	OLS	OLS	OLS	Clustered

Notes. The dependent variable is excess return at $t+1$ in decimal form. All models use the same 230 firms. Standard errors are in parentheses. Column (g) uses HC1 firm-clustered standard errors and 229 degrees of freedom for t -tests. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Task (d) asks whether the value effect differs between small and large firms. We replace continuous size with $Small_{it}$, which equals one when market capitalisation is below the full-sample median of EUR 44.917 million and zero otherwise. This common threshold is calculated across all firm-year observations, so a firm can change classification over time. Table 4 reports the annual counts.

Table 4. Small-firm observations by year

Year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
N	116	118	117	120	118	117	109	110	105	105	99	98

Notes. Small firms have market capitalisation below the full-sample median of EUR 44.917 million.

The dummy captures a difference in return levels, while its interaction with book-to-market allows a different slope for small firms. We estimate

$$r_{i,t+1}^e = \alpha + \beta_B B2M_{it} + \delta Small_{it} + \gamma(B2M_{it} \times Small_{it}) + \beta_L Lev_{it} + \beta_P Prof_{it} + \beta_D DY_{it} + \varepsilon_{it}. \quad (2)$$

The book-to-market slope is β_B for large firms and $\beta_B + \gamma$ for small firms. The null hypothesis $H_0: \gamma = 0$ therefore states that the slopes are equal, while $H_1: \gamma \neq 0$ states that they differ. We reject the null ($t = 3.252$, $p = 0.001$). A one-unit higher book-to-market ratio is associated with returns 4.259 percentage points higher among large firms and 8.874 points higher among small firms. The difference of 4.615 points indicates an economically stronger value effect among small firms, although the regression cannot distinguish risk compensation from mispricing.

Having established a difference between the slopes, task (e) tests whether book-to-market is unrelated to returns in both groups. This requires the joint null $H_0: \beta_B = \gamma = 0$, against $H_1: \beta_B \neq 0$ or $\gamma \neq 0$. Comparing equation (2) with a restricted model excluding both book-to-market terms gives $F(2, 2657) = 47.512$ ($p < 0.001$). We reject the joint null. This agrees with the positive relationship in model (c), which imposed one common book-to-market slope. The interaction model shows that this common slope conceals a stronger relationship among small firms. The restricted model and test are included in the Appendix.

Differences across calendar years and industries could also influence the estimated relationship. Model (f) therefore adds year effects λ_t and two-digit SIC industry effects $\eta_{s(i,t)}$,

$$r_{i,t+1}^e = \alpha + \beta_B B2M_{it} + \delta Small_{it} + \gamma(B2M_{it} \times Small_{it}) + \beta_L Lev_{it} + \beta_P Prof_{it} + \beta_D DY_{it} + \lambda_t + \eta_{s(i,t)} + \varepsilon_{it}. \quad (3)$$

Year effects capture common annual shocks, while industry effects capture average return differences across industries. The model includes 11 year dummies and nine industry dummies, with 2010 and SIC 20 as the respective reference categories. Both book-to-market coefficients remain close to column (d)/(e) and significant at 1%. Adjusted R^2 rises from 0.097 to 0.343, and the added dummies are jointly significant ($F(20, 2637) = 50.607, p < 0.001$). This supports the richer specification because it explains more return variation even after accounting for the additional regressors. It does not, however, establish a causal effect of valuation.

Finally, column (g) uses HC1 standard errors clustered by firm. This allows the error variance to differ across observations and errors for the same firm to be correlated over time. Coefficients and model fit are unchanged because clustering changes only the estimated uncertainty. Both book-to-market terms remain significant at 1%. Profitability is positive and significant ($\hat{\beta}_P = 0.131, t = 2.335, p = 0.020$). Using the unrounded estimate, a ten-percentage-point increase in profitability corresponds to $\hat{\beta}_P \times 0.100 \times 100 \approx 1.311$ percentage points higher subsequent excess returns, holding the other regressors fixed.

Event Study

We now turn from annual return differences to the market reaction around a specific event. The event date is Friday, 24 June 2016, when the Brexit referendum result became known before trading. The $[-1, +1]$ window covers 23, 24 and 27 June, and $[-2, +2]$ covers 22, 23, 24, 27 and 28 June. Both exclude the intervening weekend.

Using 2015 classifications, UK sales exposure means that at least 10% of sales originate in the UK. The alternative treatment is a UK listing. The final panel contains 54 sales-exposed and 176 control firms, compared with 40 UK-listed and 190 control firms. Table 5 shows that the definitions disagree for 74 firms (32.174%).

Table 5. UK exposure classifications in the final sample

UK sales share	Not UK-listed	UK-listed	Total
Below 10%	146	30	176
At least 10%	44	10	54
Total	190	40	230
Classified differently			74

We use the supplied group-level average abnormal returns and standard errors. Cumulative abnormal returns sum daily abnormal returns. Their standard errors equal the daily standard error times $\sqrt{L}$ for L trading days, assuming no correlation across days. For independent treated and control groups, $SE_{\text{diff}} = \sqrt{SE_T^2 + SE_C^2}$. Test statistics divide each return or difference by its standard error. Table 6 reports two-sided normal-approximation tests under both definitions.

Table 6. Abnormal returns around the Brexit referendum result

Definition	Window	Treated	Control	Difference
UK sales	Event day	-4.678*** (-28.682)	-0.032 (-0.371)	-4.646*** (-25.108)
	[-1, +1]	-4.878*** (-17.267)	0.097 (0.638)	-4.975*** (-15.522)
	[-2, +2]	-4.927*** (-13.509)	-0.065 (-0.331)	-4.862*** (-11.752)
UK listing	Event day	-1.040*** (-5.538)	-1.121*** (-13.278)	0.081 (0.395)
	[-1, +1]	-0.949*** (-2.918)	-1.076*** (-7.359)	0.127 (0.357)
	[-2, +2]	-1.051** (-2.502)	-1.219*** (-6.453)	0.168 (0.364)

Notes. Group returns are in percent. Differences are treated minus control in percentage points. Test statistics are in parentheses. Event-day values are average abnormal returns. Longer windows report cumulative average abnormal returns. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Sales-exposed firms lose 4.678% on the event day, compared with 0.032% for controls. The treated-minus-control difference is -4.646 percentage points and remains similar over three and five days (-4.975 and -4.862 points). All differences are significant at 1%, whereas control-group returns are insignificant. Under the listing definition, both groups lose roughly 1%, but none of their differences is significant.

UK sales capture exposure to UK demand and revenues, whereas a UK listing identifies a trading venue. The conclusion about differential losses therefore depends on the exposure measure. The supplied return file has no firm identifiers, so its aggregates cannot be recalculated for the filtered panel.

Conclusion

The simulated panel displays a positive value effect that is stronger among small firms and remains significant with year and industry controls and firm-clustered standard errors. Profitability is also positively associated with subsequent returns. These patterns are consistent with familiar asset-pricing intuition, but simulated data cannot independently validate a real-world return premium. The Brexit results favour UK sales over listing status as a measure of operating exposure, showing why the economic meaning of a treatment definition matters for the conclusion.

Appendix

The complete public repository contains the reports, data, R code and pinned package versions: <https://github.com/JuliusD9500/empirical-research-in-finance-2026>.

Analyses were conducted in R. Tables were formatted separately in LaTeX.

Run the scripts from the repository root in the order below, with both supplied data files in `data/raw/`. The setup restores the versions recorded in `renv.lock`. The analysis scripts display results in R. The report tables are maintained separately in `paper/group/tables/` and are not overwritten when the analysis is run.

A.1 Environment setup

```
# =====  
# Empirical Research in Finance - initial setup  
#  
# Purpose: Restore the project packages at their saved versions.  
# Input: renv.lock (run this script from the repository root)  
# Output: Isolated project library in renv/library/  
# =====  
  
renv_installed <- requireNamespace("renv", quietly = TRUE)  
  
if (!renv_installed) { # ! is logical not  
  install.packages("renv", repos = "https://cloud.r-project.org")  
}  
  
renv::load() # use this project's isolated package library  
renv::restore(prompt = FALSE) # install the versions in renv.lock
```

A.2 Data preparation

```
# =====  
# Empirical Research in Finance - Group Assignment  
#  
# Purpose: Filter the sample, create variables and winsorize.  
# Input: data/raw/value_effect_panel.dta  
# Output: data/clean/value_effect_panel_clean.rda (final sample)  
# =====  
  
library(tidyverse) # data wrangling  
library(labelled) # variable labels  
  
# Load raw data  
data <- haven::read_dta("data/raw/value_effect_panel.dta") |>  
  haven::zap_label()  
  
# Filter data (balance sheet identity is violated; german firms only, no  
  country filter needed)  
data <- data |>  
  filter(  
    !(sic2 >= 60 & sic2 <= 69), # exclude financial firms  
    total_assets > 0,           # for 'lev' and 'prof'  
    num_shares > 0,             # for 'market_cap'  
    share_price > 0,            # for 'market_cap' and 'size'  
    complete.cases(  
      firm_id, year, sic2, excess_return,  
      total_assets, num_shares, share_price,  
      book_equity, lt_debt, debt_current, ebitda, dividends_total  
    )  
  )  
  
# Create variables  
data <- data |>  
  mutate( # keep original and add new variables  
    market_cap = share_price * num_shares,  
    b2m = book_equity / market_cap,  
    lev = (lt_debt + debt_current) / total_assets,  
    prof = ebitda / total_assets,  
    div_yield = dividends_total / market_cap,  
    size = log(market_cap)  
  )  
  
# Winsorize variables (1% in each tail, no interpolation as in the lecture)  
winsorize <- function(x) {  
  sorted_x <- sort(x)  
  n <- length(sorted_x)  
  k <- floor(0.01 * n)  
  
  lower <- sorted_x[k + 1]
```

```

upper <- sorted_x[n - k]

x[x < lower] <- lower
x[x > upper] <- upper

return(x)
}

data <- data |> # keep original variables and add winsorized ones
mutate(
  excess_return_w = winsorize(excess_return),
  b2m_w = winsorize(b2m),
  lev_w = winsorize(lev),
  prof_w = winsorize(prof),
  div_yield_w = winsorize(div_yield),
  # not size bc. logarithm weakens large variations (span is plausibly)
)

# Label variables
data <- data |>
labelled::set_variable_labels(
  firm_id      = "Firm identifier",
  year         = "Calendar year",
  sic2         = "Industry (2-digit SIC)",

  total_assets = "Total assets (EUR mn)",
  book_equity  = "Book equity (EUR mn)",
  lt_debt      = "Long-term debt (EUR mn)",
  debt_current = "Debt in current liabilities (EUR mn)",
  total_liabilities = "Total liabilities (EUR mn)",
  ebitda       = "EBITDA (EUR mn)",
  num_shares   = "Shares outstanding (mn)",
  share_price  = "Year-end share price (EUR)",
  dividends_total = "Total dividends (EUR mn)",
  total_sales  = "Net sales (EUR mn)",
  rf           = "Risk-free rate (decimal)",
  excess_return = "Excess return at t+1 (decimal)",

  uk_exposure  = "UK sales share >= 10% in 2015",
  uk_listed    = "UK listing in 2015",

  # Constructed variables
  market_cap   = "Market capitalisation (EUR mn)",
  b2m           = "Book-to-market (ratio)",
  lev           = "Leverage (ratio)",
  prof          = "Profitability (ratio)",
  div_yield     = "Dividend yield (ratio)",
  size          = "Firm size (ln market capitalisation in EUR mn)",

  # Winsorized variables

```

```
    excess_return_w = "Excess return at t+1 (winsorized, decimal)",
    b2m_w           = "Book-to-market (winsorized, ratio)",
    lev_w           = "Leverage (winsorized, ratio)",
    prof_w          = "Profitability (winsorized, ratio)",
    div_yield_w     = "Dividend yield (winsorized, ratio)"
)

# Save cleaned data
save(data, file = "data/clean/value_effect_panel_clean.rda")
```

A.3 Descriptive statistics – task (a)

```
# =====  
# Empirical Research in Finance - (a)  
#  
# Purpose: Describe the final sample, including units and sample size.  
# Input: data/clean/value_effect_panel_clean.rda from  
#        code/01_data_preparation.R  
# Output: Descriptive statistics table (displayed in R)  
# =====  
  
# %% Load required packages  
library(tidyverse) # data wrangling  
library(modelsummary) # descriptive statistics table  
  
# %% Load cleaned data  
load("data/clean/value_effect_panel_clean.rda")  
  
# %% Choose features for task  
task_a_df <- data |>  
  haven::zap_label() |>  
  transmute(  
    `Excess return`1 2` = excess_return_w * 100,  
    `Book-to-market`2` = b2m_w,  
    `Leverage`2` = lev_w * 100,  
    `Profitability`2` = prof_w * 100,  
    `Dividend yield`2` = div_yield_w * 100,  
    `Size` = size,  
    `Market capitalisation` = market_cap  
  )  
  
# Define function for percentiles  
percentile <- function(x, p) {  
  return(quantile(x, probs = p))  
}  
  
# Calculate 10th and 90th percentiles  
P10 <- function(x) percentile(x, 0.1) # reminder: Python equivalent to lambda  
  x: ...  
P90 <- function(x) percentile(x, 0.9)  
  
# Units-Column for descriptive table  
table_units <- data.frame(  
  Unit = c("%", "ratio", "%", "%", "%", "ln(EUR mn)", "EUR mn")  
)  
attr(table_units, "position") <- 2  
  
# Final sample size  
n_observations <- nrow(data)  
n_firms <- n_distinct(data$firm_id)
```

```

# Descriptive statistics table
table_a <- datasummary(
  All(task_a_df) ~ Mean + SD + Min + P10 + Median + P90 + Max,
  data = task_a_df,
  add_columns = table_units,
  fmt = 3,
  align = "llrrrrrrr",
  title = "Final-sample descriptive statistics",
  notes = paste("Observations:", n_observations, "|| Firms:", n_firms, "|| 1
    Returns refer to t+1 || 2 Winsorized variables at 1% in each tail || All
    statistics use the single final sample"),
  output = "tinytable"
)

# Display tiny table
table_a

```

A.4 Regressions and tests – tasks (b)–(g)

```
# =====  
# Empirical Research in Finance - (b)–(g)  
#  
# Purpose: Estimate the return models and test the value effect.  
# Input: data/clean/value_effect_panel_clean.rda from  
#        code/01_data_preparation.R  
# Output: Tests and tables (displayed in R)  
# =====  
  
# %% Load required packages  
library(dplyr) # data wrangling  
library(sandwich) # covariance matrix  
library(lmtest) # coefficient tests  
library(modelsummary) # regression tables  
  
# %% Load cleaned data  
load("data/clean/value_effect_panel_clean.rda") # name: 'data'  
  
# %% (b) Pooled OLS regression  
model_b <- lm(  
  excess_return_w ~ b2m_w,  
  data = data  
)  
print(summary(model_b))  
  
# Test coefficient:  $H_0: \beta_B = 0$ ;  $H_1: \beta_B \neq 0$ . Reject  $H_0$  because  $p < 0.05$   
test_b <- coef(summary(model_b))["b2m_w", ]  
print(test_b)  
  
# %% (c) Add controls  
model_c <- lm(  
  excess_return_w ~ b2m_w + size + lev_w + prof_w + div_yield_w,  
  data = data  
)  
print(summary(model_c))  
  
# Test controls:  $H_0: \beta_j = 0$ ;  $H_1: \beta_j \neq 0$   
test_c <- coef(summary(model_c))[c("size", "lev_w", "prof_w", "div_yield_w"), ]  
print(test_c)  
  
# %% (d) Small firms and the value effect  
data <- data |>  
  dplyr::mutate(small_firm = as.numeric(market_cap < median(market_cap)))  
  
# Estimated model with interaction term  
model_d <- lm(  
  excess_return_w ~ b2m_w * small_firm + lev_w + prof_w + div_yield_w,
```

```

    data = data
)
print(summary(model_d))

# Test interaction:  $H_0: \gamma = 0$ ;  $H_1: \gamma \neq 0$ 
test_d <- coef(summary(model_d))["b2m_w:small_firm", ]
print(test_d)

# Annual counts table
table_d <- datasummary(
  Factor(year, name = "Year") ~ N, #left: rows, right: column -> number of
    observations
  data = data |> dplyr::filter(small_firm == 1),
  fmt = 0,
  align = "lr",
  title = "Small-firm observations by year",
  notes = "Small if market capitalisation is below the full-sample median",
  output = "tinytable"
)

# Display table
table_d

# Book-to-market slopes by firm size (percentage points)
slopes_d_pp <- c(
  large = coef(model_d)[["b2m_w"]],
  small = coef(model_d)[["b2m_w"]] + coef(model_d)[["b2m_w:small_firm"]],
  diff = coef(model_d)[["b2m_w:small_firm"]]
) * 100 # economic interpretation
print(slopes_d_pp)

# %% (e) Estimate the model without interaction term
model_e <- lm(
  excess_return_w ~ small_firm + lev_w + prof_w + div_yield_w,
  data = data
)

# Joint F-test of no value effect (value companys: high b2m + high dividend /
  growth companies: low b2m + low dividend)
#  $H_0: \beta_B = \gamma = 0$ ;  $H_1$ : at least one is nonzero.
test_e <- anova(model_e, model_d)
print(test_e)

# %% (f) Add year and industry dummies
model_f <- lm(
  excess_return_w ~ b2m_w * small_firm + lev_w + prof_w + div_yield_w +
    factor(year) + factor(sic2),
  data = data
)

```

```

print(summary(model_f))

# Test dummies: H0: all added coefficients = 0; H1: at least one is nonzero.
test_f <- anova(model_d, model_f)
print(test_f)

# %% (g) Firm-clustered standard errors
n_firms <- n_distinct(data$firm_id)
vcov_g <- vcovCL(model_f, cluster = ~ firm_id, type = "HC1")
model_g <- coeftest(model_f, vcov. = vcov_g, df = n_firms - 1)

# Test profitability: H0:  $\beta_P = 0$ ; H1:  $\beta_P \neq 0$ .
test_g <- model_g["prof_w", ]
print(test_g)

# Return change for a 10-percentage-point increase in profitability
  (percentage points)
profitability_effect_pp <- coef(model_f)["prof_w"] * 0.10 * 100
print(profitability_effect_pp)

# %% Regression table
# Combine clustered inference from (g) with model fit from (f)
model_g_table <- modelsummary(model_g, output = "modelsummary_list")
model_g_table$glance <- get_gof(model_f)

# Additional rows below the regression results
table_details <- data.frame(
  term = c("Year dummies", "Industry dummies", "Standard errors", "Firms"),
  "(b)" = c("No", "No", "OLS", n_firms),
  "(c)" = c("No", "No", "OLS", n_firms),
  "(d)/(e)" = c("No", "No", "OLS", n_firms),
  "(f)" = c("Yes", "Yes", "OLS", n_firms),
  "(g)" = c("Yes", "Yes", "Clustered", n_firms),
  check.names = FALSE
)

# Consolidated regression table; (e) tests the model from (d)
table_b_g <- modelsummary(
  list("(b)" = model_b, "(c)" = model_c, "(d)/(e)" = model_d,
       "(f)" = model_f, "(g)" = model_g_table),
  coef_map = c( # select, label and order coefficients; omit year/industry
                dummies
                "b2m_w" = "Book-to-market",
                "size" = "Size",
                "small_firm" = "Small firm",
                "b2m_w:small_firm" = "Book-to-market x Small firm",
                "lev_w" = "Leverage",
                "prof_w" = "Profitability",
                "div_yield_w" = "Dividend yield",
                "(Intercept)" = "Constant"
  )
)

```

```

),
gof_map = c("nobs", "r.squared", "adj.r.squared"),
add_rows = table_details,
statistic = "{std.error}",
stars = c("*" = 0.10, "**" = 0.05, "***" = 0.01),
fmt = 3,
align = "lrrrrr",
width = c(0.32, 0.136, 0.136, 0.136, 0.136, 0.136),
title = "One-year-ahead excess returns",
notes = paste(
  "Dependent variable: excess return at t+1 (decimal). All models use the
  same final sample.",
  "Standard errors in parentheses. Column (g): HC1 firm-clustered SEs;
  t-tests use firms minus 1 degrees of freedom."
),
output = "tinytable"
)

# Display regression table
table_b_g

```

A.5 Event study – task (h)

```
# =====  
# Empirical Research in Finance - Group Assignment  
#  
# Purpose: A comparison of the stock price reaction to the Brexit referendum  
#           between UK-exposed and non-exposed German firms.  
# Input:  
# Output:  
# =====  
library(tidyverse)  
library(haven)  
  
#Load cleaned firm panel  
load("data/clean/value_effect_panel_clean.rda")  
  
#Load event-study summary file (no cleaning needed)  
cars <- haven::read_dta("data/raw/cars_summary.dta") |> haven::zap_label()  
  
# 1. Event design  
  
event_date <- as.Date("2016-06-24") # Brexit announcement  
event_windows <- list(  
  "[-1,+1]" = c(start = as.Date("2016-06-23"), end = as.Date("2016-06-27")),  
  "[-2,+2]" = c(start = as.Date("2016-06-22"), end = as.Date("2016-06-28"))  
)  
  
# The event date is Friday, 24 June 2016. The [-1,+1] window covers  
# 23, 24 and 27 June; [-2,+2] also includes 22 and 28 June.  
# The weekend of 25-26 June is excluded from both trading-day windows.  
  
# 2. Sample composition  
  
# sanity check: uk_exposure / uk_listed should be fixed per firm  
data |>  
  group_by(firm_id) |>  
  summarise(n_exp = n_distinct(uk_exposure), n_listed = n_distinct(uk_listed))  
  |>  
  filter(n_exp > 1 | n_listed > 1) # should return 0 rows  
  
firm_level <- data |>  
  distinct(firm_id, uk_exposure, uk_listed)  
  
# counts under each definition  
firm_level |> count(uk_exposure)  
firm_level |> count(uk_listed)  
  
# cross-tabulation  
cross_tab <- firm_level |>  
  count(uk_exposure, uk_listed) |>
```

```

pivot_wider(names_from = uk_listed, values_from = n, values_fill = 0)
cross_tab

# firms classified differently across the two definitions
n_different <- firm_level |>
  filter(uk_exposure != uk_listed) |>
  nrow()
n_different

# Results explained

# Final sample: 230 firms (2664 firm-year observations)
#
# Treated/control counts under each definition:
# UK sales exposure: ( dummy variable -> if the firm gets 10% of its sales
  from the UK. 1 = yes (exposed), 0 = no.): 54 treated (1), 176 control (0)
# UK listing status: ( dummy variable -> the firm's shares are also traded
  on a UK stock exchange. 1 = yes (listed), 0 = no.) 40 treated (1), 190
  control (0)
# 4 possible pairings: (0,0), (0,1), (1,0), (1,1)

# Cross-tabulation (UK sales exposure x UK listing status):
#
#
# UK sales exp < 10% (0)      UK not listed (0)      UK listed (1)      Total
# UK sales exp >= 10% (1)      146 (0,0)          30 (0,1)          176
# Total                        44 (1,0)          10 (1,1)          54
#                               190                40            230

# The top-left cell with the number 146 means that 146 firms have less than
  10% UK sales and are not UK-listed -> firms with no UK connection under
  either measure.
# The bottom-right cell with the number 10 means that 10 firms have both 10%
  UK sales and are UK-listed -> firms connected to the UK under both
  measures.
# 74 (30+44) (32% of the sample)
# 30 firms are UK-listed but do not meet the 10% sales-exposure threshold
# 44 firms have >=10% UK sales exposure but are not UK-listed.
# This indicates the two definitions disagree.
# aspects of a firm's UK relationship (revenue exposure vs.listing venue),
  not the same underlying concept.

# (3) Statistical tests -- UK sales exposure

# CAR = cumulative abnormal return, SE = standard error,
# t-stat = CAR / SE, sig = significance stars based on the t-stat

windows <- list(
  "Event day" = 0,
  "[-1,+1]" = -1:1,
  "[-2,+2]" = -2:2

```

```

)

# sum daily AARs into CAR, scale SE by sqrt(window length), and get
# treated/control/difference. for the difference row we add variances
# (not SEs directly) since treated and control are independent groups
car_table_exposure <- map_dfr(names(windows), function(win_name) {
  win_days <- windows[[win_name]]

  cars |>
    filter(def == "uk_exposure", evttime %in% win_days) |>
    group_by(group) |>
    summarise(
      CAR = sum(aar_m),
      SE = unique(aar_se) * sqrt(length(win_days)),
      .groups = "drop"
    ) |>
    pivot_wider(names_from = group, values_from = c(CAR, SE)) |>
    mutate(
      CAR_difference = CAR_treated - CAR_control,
      SE_difference = sqrt(SE_treated^2 + SE_control^2),
      window = win_name
    )
}) |>
pivot_longer(
  cols = c(CAR_treated, CAR_control, CAR_difference, SE_treated, SE_control,
    SE_difference),
  names_to = c(".value", "group"),
  names_pattern = "(CAR|SE)_(.*)"
) |>
mutate(
  t_stat = CAR / SE,
  sig = case_when(
    2 * (1 - pnorm(abs(t_stat))) < 0.01 ~ "***",
    2 * (1 - pnorm(abs(t_stat))) < 0.05 ~ "**",
    2 * (1 - pnorm(abs(t_stat))) < 0.10 ~ "*",
    TRUE ~ ""
  )
) |>
select(window, group, CAR, SE, t_stat, sig)

car_table_exposure

# results:
# treated firms drop around 4.7-4.9% in every window and it's super
# significant (t around -13 to -29). control firms barely move at all
# and aren't significant. the difference between the two groups is
# also huge and significant (t around -12 to -25), so UK sales
# exposure really does seem to explain a big chunk of the reaction

# =====

```

```

# (4) Robustness check -- UK listing status
# =====
# same test as (3), just swapping in the UK listing dummy instead
car_table_listed <- map_dfr(names(windows), function(win_name) {
  win_days <- windows[[win_name]]

  cars |>
    filter(def == "uk_listed", evttime %in% win_days) |>
    group_by(group) |>
    summarise(
      CAR = sum(aar_m),
      SE = unique(aar_se) * sqrt(length(win_days)),
      .groups = "drop"
    ) |>
    pivot_wider(names_from = group, values_from = c(CAR, SE)) |>
    mutate(
      CAR_difference = CAR_treated - CAR_control,
      SE_difference = sqrt(SE_treated^2 + SE_control^2),
      window = win_name
    )
}) |>
  pivot_longer(
    cols = c(CAR_treated, CAR_control, CAR_difference, SE_treated, SE_control,
              SE_difference),
    names_to = c(".value", "group"),
    names_pattern = "(CAR|SE)_(.*)"
  ) |>
  mutate(
    t_stat = CAR / SE,
    sig = case_when(
      2 * (1 - pnorm(abs(t_stat))) < 0.01 ~ "***",
      2 * (1 - pnorm(abs(t_stat))) < 0.05 ~ "**",
      2 * (1 - pnorm(abs(t_stat))) < 0.10 ~ "*",
      TRUE ~ ""
    )
  ) |>
  select(window, group, CAR, SE, t_stat, sig)

car_table_listed

# results:
# this time both treated AND control firms drop by about the same
# amount (around 1-1.2%), and both are significant on their own. but
# the difference between them is tiny and never significant (t only
# around 0.36-0.40). so being UK-listed doesn't really separate firms
# that got hit harder from ones that didn't -- unlike sales exposure,
# this definition doesn't seem to capture real Brexit exposure

```

Empirical Research in Finance

Combined Course Submission · Group A

Humboldt-Universität zu Berlin
School of Business and Economics
Summer Semester 2026

PART II · INDIVIDUAL SUBMISSION · RESEARCH PROPOSAL

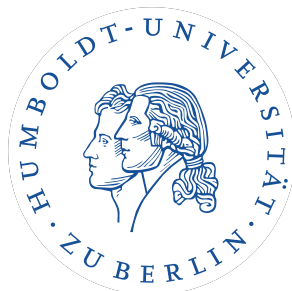
Weather Shocks and Corporate Innovation

R&D Investment and Precautionary Cash Holdings
after Hurricane Helene

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Bachelor of Science
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Course	Reference	Instructor	Submission
Empirical Research in Finance	701143	Dr. Beatriz Mariano	15 September 2026



Background

Hurricane Helene caused flooding and wind damage across parts of the southeastern United States in late September 2024. Such a shock can force firms to balance long-term investment against immediate funding needs. Research and development (R&D) is particularly relevant because its benefits often arrive well after the initial spending.

Dessaint and Matray (2017) find that firms near a hurricane disaster area, but not directly affected, temporarily increase cash holdings and discuss hurricane risk more often in regulatory filings. They interpret this as salience: a nearby disaster makes liquidity risk more prominent in managers' decisions. Their study motivates examining investment alongside cash, although directly exposed firms also face damage and operational disruption.

The research question is: how does R&D investment change after Helene among firms headquartered in affected counties, relative to firms elsewhere, and is a decline accompanied by higher cash holdings? The joint pattern would be consistent with greater emphasis on liquidity, but would not by itself establish a precautionary motive.

Expected Outcome

My main hypothesis (H_1) is that firms headquartered in affected counties reduce R&D investment after Helene relative to other firms. I also expect a relative increase in cash holdings, which would provide complementary evidence about their financial response.

The rationale is that uncertain future revenues and urgent repair costs may make firms more reluctant to commit funds to new research projects. R&D often creates intangible assets that are difficult to pledge as collateral. Cash, in contrast, can cover several immediate needs. Managers may therefore postpone new projects or research hiring while preserving liquidity.

The prediction is not certain. Established research programmes can be costly to interrupt, and affected firms may spend existing cash on repairs. This makes the direction of both outcomes an empirical question rather than a necessary consequence of the storm.

Data

I would use a quarterly panel of US-domiciled, US-listed firms from 2022Q1 to 2025Q4. Restricting firms to December fiscal year ends aligns their reporting periods. I would exclude 2024Q3, when Helene occurred, leaving ten pre-storm quarters and five full post-storm quarters. Quarterly data separate the periods before and after the shock and allow me to examine whether any response persists. Starting in 2022 avoids the initial pandemic disruption, although later macroeconomic shocks still need to be considered.

Accounting data would come from Compustat North America through WRDS, subject to obtaining access. R&D investment would equal 100 times quarterly research and development expense ($XRDQ$) divided by total assets (ATQ) at the end of 2023. Cash holdings would equal 100 times cash and short-term investments ($CHEQ$) divided by the same fixed asset base. Fixing the denominator prevents changes in reported assets after Helene from mechanically changing either ratio. Both outcomes are measured in percentage points of pre-storm assets.

I would focus on firms reporting R&D before Helene and exclude financials (SIC 6000 to 6999) and utilities (SIC 4900 to 4999). Missing R&D would remain missing unless company filings explicitly establish a zero value. I would require positive assets at the end of 2023 and compare both outcomes on observations where each is available. Firms that later leave the market would remain for available periods. Extreme ratios would be checked against filings, with sensitivity to winsorisation at the 1st and 99th percentiles reported.

Exposure would be based on FEMA's county designations for Individual Assistance following Helene. E_i would equal one if a firm's pre-storm headquarters lies in a designated county, and zero otherwise. I would verify historical headquarters addresses using SEC filings rather than assume that current Compustat addresses were valid in 2024. Other mainland firms would form the comparison group, after checking industry composition and prior outcome trends. This definition measures headquarters exposure and does not establish damage to every firm's operations.

Methodology

I would estimate a difference-in-differences model by OLS to compare changes in exposed and unexposed firms:

$$Y_{it} = \alpha_i + \lambda_t + \beta(E_i \times \text{Post}_t) + \varepsilon_{it}.$$

Here, i identifies firms and t identifies quarters. I would run the regression separately for R&D investment and cash holdings, as defined above. Firm fixed effects (α_i) capture persistent differences between firms. Quarter fixed effects (λ_t) capture shocks shared across firms in a quarter, and ε_{it} is the error term. Standard errors would be clustered by firm because observations from the same firm may be correlated over time.

Post_t equals one from 2024Q4 onwards and zero before Helene. The individual exposure and post-event dummies are absorbed by the fixed effects. The coefficient β measures the change in exposed firms relative to unexposed firms, in percentage points of pre-storm assets.

For R&D, I would test $H_0 : \beta \geq 0$ against $H_1 : \beta < 0$. The complementary cash test is $H_0 : \beta \leq 0$ against $H_1 : \beta > 0$. I would use one-sided t-tests and report 95% confidence intervals. The interpretation assumes that both groups would have followed parallel trends without Helene. Plots of pre-storm group means would help assess this assumption. A fall in R&D alongside higher cash would fit the proposed liquidity response, while leaving its underlying motive unresolved.

Extensions and Limitations

One limitation is that headquarters may be far from laboratories and other operating sites. Exposure can therefore be misclassified, and the resulting bias need not always reduce the estimated effect. Local demand changes, damage and financing decisions may also affect both outcomes. Their joint movement alone cannot identify precautionary behaviour.

As an extension, I would examine whether responses differ with initial financing capacity. A dummy would identify firms with end-2023 assets below the sample median and debt/assets above the median. This is a proxy for financing constraints, not a direct measure. I would add its interaction with $E_i \times \text{Post}_t$, together with all lower-order terms not absorbed by fixed effects. A more negative R&D response for this group would be consistent with financing frictions, although it would not establish the cash-preservation mechanism.

Reference

Dessaint, O. and Matray, A. (2017). Do managers overreact to salient risks? Evidence from hurricane strikes. *Journal of Financial Economics*, 126(1), 97–121.
<https://doi.org/10.1016/j.jfineco.2017.07.002>

Appendix

For transparency, this appendix lists, in chronological order, the prompts submitted by the author to Claude during the preparation of this proposal. Prompts are reproduced verbatim, including original wording and typographical errors.

Prompt 1

“can you tell me the economic damage of the hurricane helene in the states and which areas it affected the most. i would also like to have an info according the insurances that companies and houses had at that time and in these areas that were affected”

Prompt 2

“using these recommended journals Journal of Finance I Journal of Financial Economics I Review of Financial Studies I Journal of Financial and Quantitative Analysis I Review of Finance I Journal of Financial Intermediation I Journal of Corporate Finance I Management Science I Journal of Money, Credit and Banking I Review of Corporate Finance Studies I Review of Asset Pricing Studies I Journal of Business I Management Science I Journal of Banking and Finance I Journal of Accounting and Economics “Good” journals This is the list of the best Economics journals, based on journal rankings. I American Economic Review I Journal of Political Economy I Quarterly Journal of Economics I Econometrica I Review of Economic Studies I American Economic Journal (Applied Economics, Economic Policy, Macroeconomics, Microeconomics) I Journal of Monetary Economics I RAND Journal of Economics I Journal of the European Economic Association, can you give me a reference that fits the most with my research question, which is Hurricane Strikes and Innovation: Does Direct Exposure to Hurricane Helene Reduce Corporate R&D and Increase Precautionary Cash Holdings?”

Prompt 3

“which percent was approximately insured in the affected areas”

Prompt 4

“is r&d risky ?”

Prompt 5

“according to my question would you suggest to use yearly firm panel or quarterly firm panel which one would be more useful”

Prompt 6

“hmm i dotn really agree The point about quarterly R&D being noisy is really good and will impress the marker. But if you use quarterly for cash and yearly for R&D, then you need two different datasets, two different Post definitions, and two regressions in your Methodology. For a 3-page proposal with one regression model, that gets messy fast.

The safer compromise is: keep one quarterly panel as your main sample for both, and mention yearly R&D only as a robustness check. You keep the insight but your Methodology stays clean.”

Prompt 7

“how is r&d intensity measured”

Prompt 8

“how did they divide counties were hit badly enough by Helene and the ones that didnt how did they measure and name the treated and the control firms”

Prompt 9

“would difference in differences be good to compare treated and control firms before and after Helene”

Prompt 10

“fix the grammatical mistakes and the structure”